# Linear Algebra — Lecture 2

Field, Vector Space, Subspace, Linear Independence, Basis and Dimension

Friedberg–Insel–Spence, *Linear Algebra* 4e, §1.2, §1.3, §1.5, §1.6
D. C. Lay, *Linear Algebra and Its Applications* 4e, §1.7, §4.1–§4.5

**Topics.** Field, vector space, subspace, linear independence, basis and dimension.

**How the two texts divide the work.** FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the coordinate intuition. Mapping for this lecture:

| Topic | Lay 4e (printed) | Friedberg 4e (printed) |
|---|---|---|
| Field, vector space | §4.1 (p. 190) | §1.2 (p. 6) |
| Subspace | §4.1 (p. 193), §4.2 (p. 198) | §1.3 (p. 16) |
| Linear independence | §1.7 (p. 55), §4.3 (p. 208) | §1.5 (p. 35) |
| Basis | §4.3 (p. 209), §4.4 (p. 216) | §1.6 (p. 42) |
| Dimension | §4.5 (p. 225) | §1.6 (p. 42–57) |

**Two questions that drive the whole lecture:** (1) *Independence*: is there a redundant vector in the set? (2) *Spanning*: does the set have enough vectors to reach every vector of the space?

*How to read:* use Lay's concrete examples in $\mathbb{R}^n$ to build the picture, then Friedberg's axiomatic proofs to see why each test is legitimate.

*All page numbers here are printed pages.* For Lay, PDF page = printed page +17; for Friedberg, PDF page = printed page +15.

## 1 Fields and Vector Spaces

**Definition 1.1** (Field)**.** A **field** is a set $\mathbb{F}$ with two operations, addition and multiplication, such that addition is commutative and associative with an additive identity and additive inverses, multiplication is commutative and associative with a multiplicative identity and inverses for nonzero elements, and multiplication distributes over addition.

**Definition 1.2** (Vector space; Friedberg, §1.2, p. 6–7)**.** Let $\mathbb{F}$ be a field. A **vector space** (or linear space) over $\mathbb{F}$ is a nonempty set $V$ with two operations, **addition** and **scalar multiplication**, satisfying the following axioms for all $x, y, z \in V$ and all $a, b \in \mathbb{F}$:

(VS 1) $x + y = y + x$ (commutativity);
(VS 2) $(x + y) + z = x + (y + z)$ (associativity);
(VS 3) $0$ with $x + 0 = x$ (zero vector);
(VS 4) for each $x$ a $y$ with $x + y = 0$ (negative);
(VS 5) $1x = x$ (identity $1 \in \mathbb{F}$);
(VS 6) $(ab)x = a(bx)$;
(VS 7) $a(x + y) = ax + ay$;
(VS 8) $(a + b)x = ax + bx$.

**Remark 1.1** (The two books list the axioms differently, but equivalently)**.** Lay (§4.1, p. 190) defines a vector space as a nonempty set with two operations satisfying **ten** axioms. His ten axioms are the eight above **plus two closure requirements** ($x + y \in V$ and $ax \in V$, which he numbers Axiom 1 and Axiom 6).

The two formulations are equivalent: saying that addition is *an operation on* $V$ already presupposes closure. Either formulation is fine to use, but cite it precisely – Lay's Axiom 3 or Friedberg's (VS 3).

**Proposition 1.1** (Three facts that follow from the axioms; Lay, §4.1, p. 191, equations (1)–(3); Friedberg leaves these as exercises)**.** *For every $x \in V$ and $a \in \mathbb{F}$,*

$$0x = \mathbf{0}, \qquad a\mathbf{0} = \mathbf{0}, \qquad (-1)x = -x.$$

*Proof.* All three are proved the same way: split 0 as $0+0$ (or read the distributive law backwards), then cancel what is left using the zero vector and additive inverses.

(i) $0x = \mathbf{0}$. By (VS 8) and (VS 3),

$$0x + 0x = (0+0)x = 0x = 0x + \mathbf{0}.$$

Adding $-(0x)$ to both sides gives $0x = \mathbf{0}$.

(ii) $a\mathbf{0} = \mathbf{0}$. Likewise $a\mathbf{0} + a\mathbf{0} = a(\mathbf{0}+\mathbf{0}) = a\mathbf{0} = a\mathbf{0} + \mathbf{0}$, so $a\mathbf{0} = \mathbf{0}$.

(iii) $(-1)x = -x$. By (VS 8) and (i),

$$x + (-1)x = (1 + (-1))x = 0x = \mathbf{0}.$$

The vector $y$ with $x + y = \mathbf{0}$ is unique (uniqueness of the zero vector and of inverses, Friedberg, Theorem 1.1, p. 11), so $(-1)x = -x$. □

**Remark 1.2** (Examples: vector spaces are far more general than $\mathbb{R}^n$)**.**

- $\mathbb{R}^n$ for every $n \geq 1$; more generally $\mathbb{F}^n$. Lay Example 1, p. 191.
- All directed line segments in space (equal when same length and direction), with the parallelogram rule. Lay Example 2, p. 191.
- All doubly infinite sequences $\{y_k\} = (\dots, y_{-1}, y_0, y_1, \dots)$. Lay Example 3, p. 191.
- $\mathcal{F}(S, \mathbb{F})$: for a nonempty set $S$ and a field $\mathbb{F}$, **all functions** from $S$ to $\mathbb{F}$, with addition and scalar multiplication defined **pointwise**: $(f+g)(s) = f(s) + g(s)$ and $(cf)(s) = c\,f(s)$. Friedberg Example 3, §1.2, p. 9.
- $P(\mathbb{F})$: **all** polynomials with coefficients in $\mathbb{F}$, of any degree. Lay §4.1, p. 193; Friedberg Example 4, §1.2, p. 10.
- $P_n(\mathbb{F})$: polynomials of degree at most $n$ with coefficients in $\mathbb{F}$. Lay §4.1, p. 192; Friedberg Example 5, §1.2, p. 11.
- $M_{m\times n}(\mathbb{F})$: $m \times n$ matrices with the usual operations. Friedberg Example 2, §1.2, p. 9.

The point: a "vector" need not be an arrow or a tuple; only the axioms matter.

## 2 Subspaces

**Definition 2.1** (Subspace; Lay, §4.1, p. 193, parts a–c; Friedberg, Theorem 1.3, p. 17)**.** Let $W$ be a nonempty subset of a vector space $V$. Then $W$ is a **subspace** of $V$ if $W$ is itself a vector space under the operations of $V$. Equivalently, only three conditions need checking:

(a) $\mathbf{0} \in W$;
(b) $x + y \in W$ whenever $x, y \in W$ (closed under addition);
(c) $ax \in W$ whenever $a \in \mathbb{F}$ and $x \in W$ (closed under scalar multiplication).

**Theorem 2.1** (Criterion for a subspace; Friedberg, Theorem 1.3, §1.3, p. 17)**.** *Let $V$ be a vector space and $W \subseteq V$. Then $W$ is a subspace of $V$ if and only if conditions (a), (b), (c) above hold for the operations defined on $V$.*

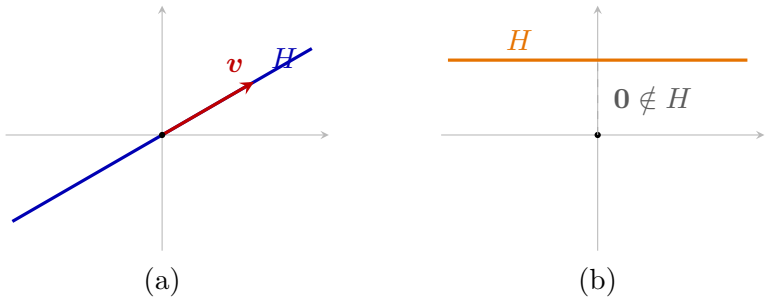
*Proof.* ($\Rightarrow$) If $W$ is a subspace then $W$ is a vector space under the operations of $V$, so (b) and (c) hold. Also $W$ has its own zero $0'$, that is, $x + 0' = x$ for every $x \in W$; but $x + 0 = x$ holds as well, so $0' = 0$ by uniqueness of the zero vector (Friedberg, Theorem 1.1, p. 11) and (a) holds.

($\Leftarrow$) Assume (a), (b), (c). By (b) and (c) $W$ is closed under both operations, so they restrict to $W$. Of the eight axioms, commutativity, associativity, the unit law $1x = x$, $a(bx) = (ab)x$ and the two distributive laws hold for every vector of $V$ and hence for every vector of $W$; (a) supplies the zero axiom. Only the additive inverse is left: if $x \in W$ then $(-1)x \in W$ by (c), and $(-1)x = -x$ (Proposition 1.1), so $-x \in W$. Hence $W$ satisfies all eight axioms, is a vector space, and therefore a subspace of $V$. □

**Theorem 2.2** (Intersections of subspaces; Friedberg, Theorem 1.4, §1.3, p. 19)**.** *Any intersection of subspaces is again a subspace: if $W_1$ and $W_2$ are subspaces of $V$, then so is $W_1 \cap W_2$.*

**Remark 2.1** (The union of subspaces is usually not a subspace; Friedberg, §1.3, p. 19)**.** A union of subspaces contains the zero vector and is closed under scalar multiplication, but closure under addition can fail: two distinct lines through the origin in $\mathbb{R}^2$ are subspaces, yet their union is not.

**Remark 2.2** (Geometric picture)**.** In $\mathbb{R}^3$, every line through the origin and every plane through the origin is a subspace. Failing to pass through the origin is enough to destroy the property, because then the zero vector is missing and condition (a) fails immediately.




(a) a line through the origin is a subspace of $\mathbb{R}^3$; (b) a line not through the origin is not a subspace: it does not contain the zero vector.

**Theorem 2.3** (A span is a subspace; Lay, Theorem 1, §4.1, p. 194)**.** *If $v_1, \ldots, v_p$ lie in a vector space $V$, then* $\operatorname{Span}\{v_1, \ldots, v_p\}$ *is a* ***subspace*** *of $V$.*

**Definition 2.2** (Null space; Lay, §4.2, p. 199)**.** Let $A$ be an $m \times n$ matrix. The **null space** of $A$, written $\operatorname{Nul} A$, is the set of all solutions of the homogeneous equation $A\boldsymbol{x} = \mathbf{0}$:

$$\operatorname{Nul} A = \{\boldsymbol{x} \in \mathbb{R}^n : A\boldsymbol{x} = \mathbf{0}\}.$$

**Theorem 2.4** (The null space is a subspace; Lay, Theorem 2, §4.2, p. 199)**.** *The null space of an $m \times n$ matrix $A$ is a subspace of $\mathbb{R}^n$.*

**Definition 2.3** (Column space; Lay, §4.2, p. 201)**.** The **column space** of an $m \times n$ matrix $A$, written $\operatorname{Col} A$, is the set of all linear combinations of the columns of $A$. If $A = [\,\boldsymbol{a}_1 \ \cdots \ \boldsymbol{a}_n\,]$, then $\operatorname{Col} A = \operatorname{Span}\{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\}$.

**Theorem 2.5** (The column space is a subspace; Lay, Theorem 3, §4.2, p. 201)**.** *The column space of an $m \times n$ matrix $A$ is a subspace of $\mathbb{R}^m$, and a typical vector in it can be written as $A\boldsymbol{x}$ for some $\boldsymbol{x}$.*

## 3 Linear Dependence and Independence

**Definition 3.1** (Linearly independent and linearly dependent; Lay, §1.7, p. 56)**.** An indexed set $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation

$$x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_p\boldsymbol{v}_p = \mathbf{0}$$

has only the trivial solution ($x_1 = \cdots = x_p = 0$), and **linearly dependent** otherwise.

**Theorem 3.1** (Characterisation of linearly dependent sets; Lay, Theorem 7, §1.7, p. 58)**.** *Let $p \geq 2$ and $\boldsymbol{v}_1 \neq \mathbf{0}$. Then $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ are linearly dependent if and only if one of them is a linear combination of the vectors before it.*

*Proof.* ($\Leftarrow$) Suppose $\boldsymbol{v}_j = c_1\boldsymbol{v}_1 + \cdots + c_{j-1}\boldsymbol{v}_{j-1}$ with $j > 1$. Then

$$c_1\boldsymbol{v}_1 + \cdots + c_{j-1}\boldsymbol{v}_{j-1} + (-1)\boldsymbol{v}_j = \mathbf{0}$$

is a relation with a *nonzero* coefficient (the one on $\boldsymbol{v}_j$), so $S$ is dependent.

($\Rightarrow$) Suppose $S$ is dependent, so $c_1\boldsymbol{v}_1 + \cdots + c_p\boldsymbol{v}_p = \mathbf{0}$ with the $c_i$ not all zero. Take the *largest* index $j$ with $c_j \neq 0$.

If $j = 1$ the relation collapses to $c_1\boldsymbol{v}_1 = \mathbf{0}$, and $\boldsymbol{v}_1 \neq \mathbf{0}$ forces $c_1 = 0$, a contradiction; hence $j > 1$. Every coefficient beyond $j$ is zero, so the relation only involves $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j$:

$$c_1\boldsymbol{v}_1 + \cdots + c_j\boldsymbol{v}_j = \mathbf{0} \quad \Longrightarrow \quad \boldsymbol{v}_j = -\frac{c_1}{c_j}\boldsymbol{v}_1 - \cdots - \frac{c_{j-1}}{c_j}\boldsymbol{v}_{j-1},$$

that is, $\boldsymbol{v}_j$ is a combination of the vectors before it. □

**Theorem 3.2** (Two quick tests; Lay, Theorem 8, §1.7, p. 59; Lay, Theorem 9, §1.7, p. 59)**.**

*(a) If a set contains **more vectors than entries in each vector**, it is linearly dependent: any set of $p > n$ vectors in $\mathbb{R}^n$ is linearly dependent.*
*(b) If a set contains the **zero vector**, it is linearly dependent.*

*Proof.* (a) Let $A = [\,\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_p\,]$. Then $A$ is $n \times p$, and $A\boldsymbol{x} = \mathbf{0}$ is a homogeneous system of $n$ equations in $p$ unknowns. If $p > n$ there are more unknowns than equations, so there are at most $n$ pivot columns and hence a free variable; therefore $A\boldsymbol{x} = \mathbf{0}$ has a nontrivial solution: $c_1\boldsymbol{v}_1 + \cdots + c_p\boldsymbol{v}_p = \mathbf{0}$ with the $c_i$ not all zero, so $S$ is dependent.

(b) Suppose $\boldsymbol{v}_k = \mathbf{0}$. Take $c_k = 1$ and every other $c_i = 0$; then $c_1\boldsymbol{v}_1 + \cdots + c_p\boldsymbol{v}_p = \mathbf{0}$ with coefficients not all zero, so $S$ is dependent.

Both are the same move – assume independence, then make it contradict itself: with $\mathbf{0}$ in the set, the "trivial solution only" that independence demands is broken by $1 \cdot \mathbf{0} = \mathbf{0}$. □

**Theorem 3.3** (Monotonicity; Friedberg, Theorem 1.6 and its corollary, §1.5, p. 39)**.** *Let $V$ be a vector space and $S_1 \subseteq S_2 \subseteq V$.*

*(a) If $S_1$ is linearly dependent, then $S_2$ is linearly dependent.*
*(b) If $S_2$ is linearly independent, then $S_1$ is linearly independent.*

**Theorem 3.4** (When can one vector be added?; Friedberg, Theorem 1.7, §1.5, p. 39)**.** *Let $S$ be a linearly independent subset of a vector space $V$ and let $v \in V$ with $v \notin S$. Then*

$$S \cup \{v\} \text{ is linearly dependent} \iff v \in \mathrm{Span}(S).$$

*Proof.* ($\Rightarrow$) Suppose $S \cup \{\boldsymbol{v}\}$ is dependent, so there are coefficients, not all zero, with

$$a\boldsymbol{v} + a_1\boldsymbol{u}_1 + \cdots + a_k\boldsymbol{u}_k = \mathbf{0}, \qquad \boldsymbol{u}_i \in S.$$

Here $a \neq 0$: if $a$ were zero we would have a nontrivial relation inside $S$, contradicting its independence. Hence

$$\boldsymbol{v} = -\frac{a_1}{a}\boldsymbol{u}_1 - \cdots - \frac{a_k}{a}\boldsymbol{u}_k \in \mathrm{Span}(S).$$

($\Leftarrow$) Suppose $\boldsymbol{v} \in \mathrm{Span}(S)$, say $\boldsymbol{v} = b_1\boldsymbol{v}_1 + \cdots + b_m\boldsymbol{v}_m$ with $\boldsymbol{v}_i \in S$. Then

$$b_1\boldsymbol{v}_1 + \cdots + b_m\boldsymbol{v}_m + (-1)\boldsymbol{v} = \mathbf{0}$$

has coefficients not all zero (the one on $\boldsymbol{v}$ is $-1$, and $\boldsymbol{v} \notin S$ makes it distinct from the $\boldsymbol{v}_i$), so $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{v}\}$ is dependent, and so is the larger set $S \cup \{\boldsymbol{v}\}$ (Theorem 3.3(a)).

This criterion is the tool the Replacement Theorem leans on: it turns "$\boldsymbol{v}$ lies in the span" into "adjoining $\boldsymbol{v}$ to an independent set destroys independence". □

# 4 Bases

**Definition 4.1** (Basis; Lay, §4.3, p. 209)**.** Let $H$ be a subspace of a vector space $V$. An indexed set $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_p\}$ in $V$ is a **basis** for $H$ if (i) $B$ is linearly independent, and (ii) $H = \operatorname{Span}\{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_p\}$.

**Theorem 4.1** (Unique representation; Lay, Theorem 7, the Unique Representation Theorem, §4.4, p. 216; Friedberg, Theorem 1.8, §1.6, p. 43)**.** *Let $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\}$ be a basis for a vector space $V$. Then for each $\boldsymbol{x} \in V$ there is a **unique** choice of scalars $c_1, \ldots, c_n$ with*

$$\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n.$$

**Theorem 4.2** (Spanning Set Theorem; Lay, Theorem 5, §4.3, p. 210)**.** *Let $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ be a set in $V$ and let $H = \operatorname{Span} S$.*

*(a) If some vector in $S$ is a linear combination of the remaining vectors, then removing it still leaves a set that spans $H$.*
*(b) If $H \neq \{\boldsymbol{0}\}$, then some subset of $S$ is a basis for $H$.*

**Theorem 4.3** (Finitely generated implies a finite basis; Friedberg, Theorem 1.9, §1.6, p. 44)**.** *If a vector space $V$ is generated by a finite set $S$, then some subset of $S$ is a basis for $V$; in particular $V$ has a finite basis.*

**Definition 4.2** (Coordinate vector; Lay, §4.4, p. 216)**.** Let $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\}$ be a basis for $V$ and let $\boldsymbol{x} \in V$. The weights $c_1, \ldots, c_n$ with $\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$ are the **coordinates of $\boldsymbol{x}$ relative to** $B$, and

$$[\boldsymbol{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

is the **coordinate vector** of $\boldsymbol{x}$ relative to $B$.

**Theorem 4.4** (The coordinate mapping is an isomorphism; Lay, Theorem 8, §4.4, p. 219)**.** *If $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\}$ is a basis for $V$, then the coordinate mapping $\boldsymbol{x} \mapsto [\boldsymbol{x}]_B$ is a **one-to-one** linear transformation from $V$ onto $\mathbb{R}^n$.*

**Theorem 4.5** (Replacement Theorem; Friedberg, Theorem 1.10, §1.6, p. 45)**.** *Let $V$ be a vector space generated by a set $G$ containing exactly $n$ vectors, and let $L$ be a linearly independent subset of $V$ containing exactly $m$ vectors. Then $m \leq n$ and there is a subset $H$ of $G$ containing exactly $n - m$ vectors such that $L \cup H$ generates $V$.*

*Proof.* By induction on $m$.

$m = 0$: here $L = \varnothing$, and $H = G$ does the job ($|H| = n = n - 0$).

Induction step: assume the theorem for some $m \geq 0$ and prove it for $m+1$. Let $L = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{m+1}\}$ be independent. By Theorem 3.3(b) the subset $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$ is independent, so the induction hypothesis gives $m \leq n$ and a subset $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-m}\}$ of $G$ such that

$$B := \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-m}\}$$

generates $V$. Hence $\boldsymbol{v}_{m+1}$ is a combination of the elements of $B$:

$$a_1\boldsymbol{v}_1 + \cdots + a_m\boldsymbol{v}_m + b_1\boldsymbol{u}_1 + \cdots + b_{n-m}\boldsymbol{u}_{n-m} = \boldsymbol{v}_{m+1}. \qquad (*)$$

If $n-m=0$, the right-hand side of $(*)$ is a combination of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$, i.e. $\boldsymbol{v}_{m+1} \in \mathrm{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$, which contradicts the independence of $L$ by Theorem 3.4. So $n-m>0$. By the same argument at least one $b_i \neq 0$ – otherwise $\boldsymbol{v}_{m+1}$ would again be a combination of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ – so assume $b_1 \neq 0$ and solve $(*)$ for $\boldsymbol{u}_1$:

$$\boldsymbol{u}_1 = -\frac{a_1}{b_1}\boldsymbol{v}_1 - \cdots - \frac{a_m}{b_1}\boldsymbol{v}_m + \frac{1}{b_1}\boldsymbol{v}_{m+1} - \frac{b_2}{b_1}\boldsymbol{u}_2 - \cdots - \frac{b_{n-m}}{b_1}\boldsymbol{u}_{n-m}.$$

Take $H = \{\boldsymbol{u}_2, \ldots, \boldsymbol{u}_{n-m}\}$. The display above says $\boldsymbol{u}_1 \in \mathrm{Span}(L \cup H)$; and

$$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{n-m} \in \mathrm{Span}(L \cup H),$$

so $B \subseteq \mathrm{Span}(L \cup H)$. Since $B$ generates $V$, $V = \mathrm{Span}(B) \subseteq \mathrm{Span}(L \cup H) \subseteq V$ (Theorem 2.3, which is Friedberg's Theorem 1.5: Span is the smallest subspace containing $L \cup H$), so $L \cup H$ generates $V$. Finally $H \subseteq G$ and $|H| = (n-m) - 1 = n - (m+1)$, which completes the induction. □

**Theorem 4.6** (Two corollaries of the Replacement Theorem; Friedberg, §1.6, Corollaries 1–2, p. 46–47)**.**

*(a) If $V$ has a finite basis, then **every basis of $V$ contains the same number of vectors**.*
*(b) Let $\dim V = n$. Any finite generating set for $V$ contains at least $n$ vectors, and a generating set containing exactly $n$ vectors is a basis.*

*Proof.* (a) Let $\beta$ be a finite basis with $n$ vectors and let $\gamma$ be any basis. If $|\gamma| > n$, apply the Replacement Theorem with $G = \beta$ ($n$ vectors) and $L$ equal to any $n+1$ vectors of $\gamma$: it gives $n+1 \leq n$, a contradiction, so $|\gamma| \leq n$. Swapping the roles ($G = \gamma$, $L = \beta$) gives $n \leq |\gamma|$. Hence $|\gamma| = n$: the size of a basis does not depend on which basis you pick, and that invariant is the **dimension**.

(b) First, "at least". Let $G$ be a finite generating set with $m$ vectors and let $\beta$ be a basis with $n$ vectors. Apply the Replacement Theorem with $G$ ($m$ vectors, generating) and $L = \beta$ ($n$ vectors, independent): $n \leq m$.

Now "exactly". Let $m = n$. By Theorem 4.3 some subset $\beta'$ of $G$ is a basis; every basis has $n$ vectors by (a), so $|\beta'| = n$; and the only $n$-element subset of an $n$-element set is the set itself, so $\beta' = G$. Thus $G$ is a basis. □

# 5 Dimension

**Definition 5.1** (Dimension; Lay, §4.5, p. 226)**.** If $V$ is spanned by a finite set, then $V$ is **finite-dimensional** and its **dimension** $\dim V$ is the number of vectors in a basis for $V$; the dimension of the zero space $\{\mathbf{0}\}$ is defined to be 0. If $V$ cannot be spanned by a finite set, it is **infinite-dimensional**.

**Theorem 5.1** (Two basic facts about dimension; Lay, Theorem 9, §4.5, p. 225; Lay, Theorem 10, §4.5, p. 226)**.**

*(a) If a basis of $V$ contains $n$ vectors, then any set in $V$ containing more than $n$ vectors must be linearly dependent.*
*(b) If $V$ has a basis of $n$ vectors, then every basis of $V$ consists of exactly $n$ vectors.*

*Proof.* These are the same facts as Theorem 4.6, phrased the way Lay states them; one application of the Replacement Theorem does it.

(a) Let $\beta$ be a basis with $n$ vectors and let $S$ have $p > n$ vectors. If $S$ were independent, the Replacement Theorem applied to the generating set $\beta$ ($n$ vectors) and $L = S$ ($p$ vectors) would give $p \le n$, contradicting $p > n$. So $S$ is dependent.

(b) This is Theorem 4.6(a): every basis of $V$ has exactly $n$ vectors. □

**Theorem 5.2** (A subspace is no larger than the whole space; Friedberg, §1.6, corollary on p. 51; Lay, Theorem 11, §4.5, p. 227; Friedberg, Theorem 1.11, §1.6, p. 50)**.** *Let $W$ be a subspace of a finite-dimensional vector space $V$. Then $W$ is also finite-dimensional and $\dim W \le \dim V$; moreover, if $\dim W = \dim V$ then $W = V$.*

*Proof.* Let $\dim V = n$.

If $W = \{\mathbf{0}\}$ then $W$ is finite-dimensional and $\dim W = 0 \le n$. Otherwise choose vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots$ in $W$ one at a time, keeping the set independent at every step. No independent subset of $V$ has more than $n$ vectors (Theorem 5.1(a)), so the process stops at some $k \le n$: $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k\}$ is independent, while adjoining any further vector of $W$ would make it dependent. By Theorem 3.4, every vector of $W$ lies in $\operatorname{Span}\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k\}$, so that set generates $W$ and is a basis of $W$; hence $\dim W = k \le n$.

If $\dim W = n$: a basis of $W$ is an independent subset of $V$ with $n$ vectors. If it did not generate $V$ there would be $\boldsymbol{v} \in V \setminus \operatorname{Span}(W)$, and by Theorem 3.4 adjoining $\boldsymbol{v}$ would give an independent set with $n+1$ vectors, contradicting Theorem 5.1(a). So it generates $V$, i.e. $W = \operatorname{Span}(W) = V$. □

**Theorem 5.3** (Basis Theorem; Lay, Theorem 12, the Basis Theorem, §4.5, p. 227)**.** *Let $V$ be a $p$-dimensional vector space with $p \ge 1$. Then*

*(a) any linearly independent set of exactly $p$ vectors in $V$ is automatically a basis for $V$;*
*(b) any set of exactly $p$ vectors that spans $V$ is automatically a basis for $V$.*

*Proof.* (a) Let $S$ have $p$ vectors and be independent. If $S$ did not generate $V$ there would be $\boldsymbol{v} \in V \setminus \operatorname{Span}(S)$; by Theorem 3.4 the set $S \cup \{\boldsymbol{v}\}$ would be independent, giving an independent set of $p+1$ vectors and contradicting Theorem 5.1(a). So $S$ generates $V$ and is a basis.

(b) Let $S$ have $p$ vectors and generate $V$. This is Theorem 4.6(b) with $n = p$: a generating set with exactly $n$ vectors is a basis. So $S$ is a basis. □

## 6 A Worked Example

**Example 6.1** (A subspace of $\mathbb{R}^3$, its basis and its dimension; This example follows the discussion in Lay, §4.3, p. 208, and Lay, §4.5, Examples 1–2)**.** Let

$$H = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathbb{R}^3 : a - 3b - c = 0 \right\}.$$

Decide whether $H$ is a subspace of $\mathbb{R}^3$, find a basis and the dimension of $H$, and decide whether $\boldsymbol{w} = (4, 0, 1)^\top$ belongs to $H$.

*Solution.* (1) Decide whether $H$ is a subspace of $\mathbb{R}^3$. Rewrite the constraint as $a = 3b + c$. Then

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3b + c \\ b \\ c \end{bmatrix} = b \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

so $H = \operatorname{Span}\left\{\begin{bmatrix}3\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}$. By Theorem 2.3 a span is a subspace, hence so is $H$.

(2) Find a basis and the dimension. The two vectors are not proportional, hence linearly independent, so they form a basis of $H$ and $\dim H = 2$.

(3) Decide whether $\boldsymbol{w} = (4, 0, 1)^\top$ lies in $H$. $4 - 3 \cdot 0 - 1 = 3 \neq 0$, so $\boldsymbol{w} \notin H$. Equivalently, with $b = 0$ and $c = 1$ we get $3b + c = 1 \neq 4$: the first coordinate 4 is not obtainable from $(3, 1)$ and $(1, 0)$. □

**Approach.** When $H$ is given by a single linear constraint, first solve the constraint for the free parameters: once $H = \operatorname{Span}\{\cdots\}$ is written down, the subspace conclusion follows immediately from the spanning-set theorem, the parameter vectors are a spanning set that one proportionality check promotes to a basis, and their number is the dimension.

# 7 Supplement: notions Homework 2 uses but the lecture did not cover

**Remark 7.1** (Where these come from)**.** None of the following is part of the lecture's main thread, yet Homework 2 uses them:

- The **sum** of subsets and the **direct sum**: the **exercises** of Friedberg, §1.3 (printed p. 22). The homework sheet supplies its own definitions (Definitions 1 and 2 there).
- **Cosets** and the **quotient space**: Friedberg, §1.3 Exercise 31 (printed p. 23). The sheet gives **no** definition, so it has to be looked up in the book; this is Homework 2 Problem 11 (20 points).
- The **trace**: Friedberg, §1.3 Example 3 (printed p. 18).
- The **matrix units** $E^{ij}$: Friedberg, §1.5 Exercise 6 (printed p. 41).
- The **direct product** $V_1 \times \cdots \times V_m$: Homework 2, Problem 5.

**Definition 7.1** (Sum of subsets, direct sum; Friedberg, §1.3 exercises, p. 22)**.** If $S_1, S_2$ are nonempty subsets of a vector space $V$, their **sum** is $S_1 + S_2 = \{x + y : x \in S_1,\ y \in S_2\}$. If $W_1, W_2$ are subspaces of $V$ with $W_1 \cap W_2 = \{0\}$ and $W_1 + W_2 = V$, then $V$ is the **direct sum** of $W_1$ and $W_2$, written $V = W_1 \oplus W_2$.

**Proposition 7.1** (Dimension of a sum; Friedberg §1.6, Exercise 29, p. 57)**.** *Let $W_1, W_2$ be finite-dimensional subspaces of a vector space $V$. Then $W_1 + W_2$ is finite-dimensional and*

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

*In particular,* $\dim(W_1 + W_2) = \dim W_1 + \dim W_2$ *if and only if* $W_1 \cap W_2 = \{\mathbf{0}\}$.

*Proof.* Take a basis $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t$ of $W_1 \cap W_2$. By the Replacement Theorem and the Basis Theorem it extends to a basis $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ of $W_1$, and also to a basis $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_q$ of $W_2$. Hence $\dim(W_1 \cap W_2) = t$, $\dim W_1 = t + p$ and $\dim W_2 = t + q$.

These $t+p+q$ vectors span $W_1+W_2$: every element of $W_1$ is a combination of $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$, every element of $W_2$ is a combination of $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_q$, and a sum of two such combinations is again one.

They are linearly independent. If $\sum_i a_i \boldsymbol{u}_i + \sum_j b_j \boldsymbol{v}_j + \sum_k c_k \boldsymbol{w}_k = \mathbf{0}$, then $\sum_k c_k \boldsymbol{w}_k$ lies in $W_2$ and equals $-\big(\sum_i a_i \boldsymbol{u}_i + \sum_j b_j \boldsymbol{v}_j\big)$, which lies in $W_1$; so it lies in $W_1 \cap W_2$ and is a combination of $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t$. But $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_q$ is a basis of $W_2$ and a vector has unique coefficients in a basis, so every $c_k = 0$. Then $\sum_i a_i \boldsymbol{u}_i + \sum_j b_j \boldsymbol{v}_j = \mathbf{0}$, and since $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ is a basis of $W_1$, all $a_i$ and $b_j$ vanish.

So these $t+p+q$ vectors form a basis of $W_1+W_2$ and $\dim(W_1+W_2) = t+p+q$. From $t+p = \dim W_1$ and $t+q = \dim W_2$ we get $t+p+q = \dim W_1 + \dim W_2 - t$, which is the claimed identity.

Finally, $\dim(W_1+W_2) = \dim W_1 + \dim W_2$ holds exactly when $t = 0$, that is, when $W_1 \cap W_2 = \{\mathbf{0}\}$. □

**Definition 7.2** (Coset and quotient space; Friedberg, §1.3 Exercise 31, p. 23)**.** Let $W$ be a subspace of $V$ and $v \in V$. The set $\{v\}+W = \{v+w : w \in W\}$ is the **coset** of $W$ containing $v$, customarily written $v+W$. The set of all cosets is $V/W$; with

$$(v_1+W)+(v_2+W) = (v_1+v_2)+W, \qquad a(v+W) = av+W,$$

it becomes a vector space, the **quotient space** of $V$ modulo $W$.

**Proposition 7.2** (Basic properties of cosets; Friedberg, §1.3 Exercise 31, p. 23)**.** *Let $W$ be a subspace of $V$ and $v, v_1, v_2 \in V$.*

*(1) $v+W$ is a subspace of $V \iff v \in W$;*
*(2) $v_1+W = v_2+W \iff v_1 - v_2 \in W$;*
*(3) the two operations above do not depend on the representatives chosen;*
*(4) $V/W$ is a vector space under those two operations.*

*Proof.* (1) If $v+W$ is a subspace, then $0 \in v+W$, so $0 = v+w$ for some $w \in W$, hence $v = -w \in W$. Conversely, if $v \in W$ then $v+W \subseteq W$ (every $v+w$ lies in $W$) and $W \subseteq v+W$ (as $w = v+(w-v)$), so $v+W = W$ is a subspace.

(2) If $v_1+W = v_2+W$, then $v_1 \in v_1+W = v_2+W$ gives $v_1 = v_2+w$ with $w \in W$, so $v_1-v_2 = w \in W$. Conversely, if $v_1 - v_2 \in W$ then $v_1 = v_2 + (v_1 - v_2) \in v_2+W$, so $v_1+W \subseteq v_2+W$; swapping the subscripts gives the reverse inclusion, and the two cosets are equal.

(3) Suppose $v_1+W = v_1'+W$ and $v_2+W = v_2'+W$. By (2), $v_1 - v_1'$ and $v_2 - v_2'$ lie in $W$, so

$$(v_1+v_2)-(v_1'+v_2') = (v_1-v_1')+(v_2-v_2') \in W, \qquad av_1 - av_1' = a(v_1-v_1') \in W,$$

and (2) again gives $(v_1+W)+(v_2+W) = (v_1'+W)+(v_2'+W)$ and $a(v_1+W) = a(v_1'+W)$.

(4) Verify the eight axioms through the representatives: $0+W$ is the zero element, $-v+W$ is the negative of $v+W$, and the remaining six follow by substituting representatives into the axioms of $V$, for instance

$$(v_1+W)+(v_2+W) = (v_1+v_2)+W = (v_2+v_1)+W = (v_2+W)+(v_1+W). \qquad \square$$

**Definition 7.3** (Trace; Friedberg, §1.3 Example 3, p. 18)**.** The **trace** of an $n \times n$ matrix $M$ is the **sum** of its diagonal entries: $\operatorname{tr}(M) = M_{11} + M_{22} + \cdots + M_{nn}$ .

**Definition 7.4** (Matrix units; Friedberg, §1.5 Exercise 6, p. 41)**.** In $M_{m\times n}(F)$ let $E^{ij}$ be the matrix whose only nonzero entry is a 1 in row $i$ and column $j$. The set $\{E^{ij} : 1 \le i \le m,\ 1 \le j \le n\}$ is linearly independent.

**Definition 7.5** (Direct product; Homework 2, Problem 5)**.** Let $V_1, \ldots, V_m$ be vector spaces over the same field $\mathbb{F}$. On the set of ordered tuples

$$V_1 \times \cdots \times V_m = \{\, (v_1, \ldots, v_m) :\ v_i \in V_i \,\}$$

define addition and scalar multiplication componentwise: $(v_1, \ldots, v_m) + (w_1, \ldots, w_m) = (v_1 + w_1, \ldots, v_m + w_m)$ and $a(v_1, \ldots, v_m) = (av_1, \ldots, av_m)$. This is a vector space over $\mathbb{F}$, called the **direct product** of the spaces. If every $V_i$ is finite-dimensional, then

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m,$$

since putting a basis of each $V_i$ in slot $i$ gives a basis of the direct product.

# 8 Checklist and Exercises

By the end of this lecture you should be able to:

1. write down the eight (or ten) axioms of a vector space and explain why the two formulations agree;
2. give an example showing that "a set with two closed operations" need not be a vector space, naming the axiom that fails;
3. decide whether a given subset of $\mathbb{R}^n$ is a subspace, and say which condition fails;
4. explain why a line not through the origin is not a subspace, using the zero vector;
5. decide whether a set of vectors is linearly dependent – both by solving and by using Theorem 3.1 and the two quick tests;
6. find a basis and the dimension of a given subspace;
7. state the Unique Representation Theorem and relate it to "basis = independent + spanning";
8. use the two corollaries of the Replacement Theorem to bound how many vectors a basis can have.

**Exercises (printed pages).** Lay: §1.7 p. 60, §4.1 p. 195, §4.2 p. 205, §4.3 p. 213, §4.4 p. 222, §4.5 p. 229; Friedberg: §1.2 p. 12, §1.3 p. 19, §1.5 p. 40, §1.6 p. 53. Each section also opens its exercises with Practice Problems; do those first.

*How to use the two texts together:*

- **Examples from Lay, theorems from Friedberg.** Lay's $\mathbb{R}^n$ examples are the most concrete; Friedberg proves everything over a general field $\mathbb{F}$ and contains the Replacement Theorem and its corollaries, which Lay does not have.
- The spine of this lecture: *axioms define the space → subspaces are tested by three conditions → a basis is "independent + spanning" → the dimension is the invariant size of a basis.*
- Lay's §1.7 treats linear independence in $\mathbb{R}^n$ and only §4.3 moves it to a general vector space. If time is short, read Theorem 3.1 (that is, Lay's Theorem 4 in §4.3) and the definition of a basis first, then go back to §1.7.